MIL-T-23593(Wep)
23 January 1963

MILITARY SPECIFICATION

FOR

TEST BENCH HARNESS, INFRARED SET
AN/AAM-4

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall facilitate C and D level maintenance of the AN/AAS-15 Detecting Set at land-based or ship-borne radar shops. Means shall be included for signal generation, monitoring, power distribution, mounting, interconnection, and test access to facilitate the maintenance by substitution method of fault isolation and repair. This equipment shall provide radar shop facilities for the AN/AAS-15 Detecting Set when complemented by the Auxiliary Test Equipment as detailed in 3.3.4 of this specification.

1.2 Classification - The Test Bench Harness, Infrared Set shall consist of the following items:

Items	Type Designation	Appl. Para.
Test Bench Harness, Infrared Set	AN/AAM-4	
Junction Box	J-2121/AAM-4	3.4.1
Rack, Electronic Equipment	MT-2808/AAM-4	3.4.2
Rack, Electronic Equipment	MT-2809/AAM-4	3.4.3
Rack, Electronic Equipment	MT-2810/AAM-4	3.4.4
Extender, Module		3.4.5
Extender, Module		3.4.5
Extender, Module		3.4.5
Wrench, Spanner	TL-692/AAM-4	3.4.6
Cable Assembly (5)	(*)	3.4.7

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-21200 Test Equipment for Use with Electronic and Fire Control Systems, General Specifications for

FSC 1290

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MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment

STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts
MS 28053	Meter, Time Totalizing, 400-Cycle
MS 3106R-36-10P	Connector, Electrical, Plug
MIL-P-116	Preservation, Method of

DRAWINGS

Bureau of Ships

5000-S9200-1385763	Table Electrical Test and Maintenance
S9101-212410	P.S.N.S. Type Plan Workbench for Aviation Electronics Shop

Hughes Aircraft Company (HAC)

486902-200	Schematic Diagram - Test Bench Harness, Infrared Set

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PUBLICATIONS

Bureau of Aeronautics

NAVAER 16-1-524 — Design Guidance Manual for Test Bench Harnesses

Bureau of Ships (Instruction)

BUSHIPS 9620.54 Ser 660-6758 — Test Bench Receptacle Panel, Type A-47 and A-48 Series for Aviation Electronics and Electrical Shops; Requested Modification of

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

(2) Nonrepairable subassemblies, as outlined in specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

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(3) When previously produced models of this equipment did not use nonrepairable sub-assemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported, as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction -

3.2.1 General - This equipment shall conform with the applicable design and construction requirements of specification MIL-T-21200 Class 3, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 150 pounds.

3.2.3 Operating Life -

3.2.3.1 Operational Stability - The equipment shall operate with specified performance for 24 hours, continuously or intermittently without the necessity for readjustment of any controls which are inaccessible during normal use.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be as specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment shall have a minimum of 200 hours of mean operating time between failures. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

3.2.3.4 Time Totalizing Meter - A time totalizing meter conforming with MS 28053 shall be included in the Test Bench Harness.

3.2.4 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200, Class 3, except as stated in the following paragraphs.

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3.2.4.1 Connectors for Use with Equipment under Test - Connectors supplied with the Test Bench Harness shall mate with the receptacles of the equipment under test.

3.2.4.2 Power Connectors - Power connectors shall conform to BUSHIPS Instruction 9260.54 Ser 660-6758, dated 27 May 1959.

3.2.4.3 Connector to Radar Test Harness - The connector used for the tie-in with the Radar Test Harness shall conform to Standard MS-3106R-36-10P.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0V 3-Phase, 400 cps and 27 ± 2V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 3 equipment, except as modified herein.

3.2.10.1 Vibration - The test equipment shall not suffer damage or fail to provide specified performance when subjected to continuous vibration along each of the three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 - 15	0.06 inch
16 - 25	0.04 inch
26 - 33	0.02 inch

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the driver of the vehicle to view, from his normal driving position, the entire

3.2.11 Warm-Up Time - The time required for the equipment to warm up prior to operations shall be kept at a minimum and shall not exceed 5 minutes under standard conditions, 15 minutes at extreme service conditions.

3.2.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-21200 for Class 3 equipment. The power required shall be of the following type(s) and shall not exceed the specified amounts.

AC Power	115V 400 cps, 3-Phase 300 VA/Phase at .8 PF Lagging
DC Power	27V 8.5 amperes

3.2.12.1 Voltage Protection - The equipment shall not be damaged by voltages either above or below the voltages specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Functional Performance - The Test Bench Harness shall be used for maintenance, repair and adjustment of the units of the Detecting Set, Infrared, AN/AAS-15, as specified in 3.3.3. It shall be capable of autonomous operation with the Detecting Set, Infrared, AN/AAS-15, or it may be operated in a system mode with the Radar Set AN/APQ-83 or AN/APQ-94 and the Radar Test Harness AN/APM-145.

3.3.3 Prime Equipment Serviced - The Test Bench Harness shall provide for mounting, interconnecting and distributing power to the following units of the Detecting Set, Infrared AN/AAS-15:

PN	AN Nomenclature
R-1075/AAS-15	Receiver, Infrared
AM-3046/AAS-15	Amplifier-Converter
PP-3047/AAS-15	Power Supply
MX-3550/AAS-15	Tank, Liquid Gas

3.3.4 Auxiliary Test Equipment - The Test Bench Harness shall include provisions for mounting, interconnecting and supplying power to the following test equipment:

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PN	Nomenclature
AN/AAM-6	Test Set, Infrared Detecting Set
AN/AAM-5	Simulator, Infrared Target
AN/USM-105	Oscilloscope or equivalent

3 3.5 Benches - The Test Bench Harness shall be so designed as to fit the Ship Shop Bulkhead Bench (BUSHIPS Drawing S9101-212410, Alt. 14) without alteration except for mounting holes in rear panels. The Test Bench Harness shall also fit two combined Shore Shop Benches (BUSHIPS Drawing 5000-S9200-1385763) suitably modified for this purpose. Information as to the alteration required for the two Shore Shop Benches shall be provided by the contractor.

3.4 Detail Requirements -

3.4.1 Junction Box - The Junction Box shall meet the following requirements:

3.4.1.1 Function - The Junction Box shall contain all necessary controls, switches, monitoring equipment and connectors. A bracket for mounting MX-3550/AAS-15 Liquid Gas Tank and a clamp for holding the Range Controller of the Target Simulator shall be fastened to the lower side of the Junction Box. The Junction Box shall be mounted either under the top shelf or the rear panel of the Ship Shop Bulkhead Bench. It shall also be capable of being mounted under the top shelves and/or rear panels of two suitably modified Shore Shop Benches.

3.4.1.2 Form Factor - The Junction Box shall not exceed 48 inches in width, 11 inches in height, and 16 inches in depth. These dimensions do not include the bracket for the Liquid Nitrogen Tank nor the clamp for the Range Controller of the Target Simulator.

3.4.1.3 Contents of Unit - The Junction Box shall contain the following assemblies, subassemblies and circuits.

3.4.1.3.1 Chassis - Two removable chassis having front panels approximately 19 inches wide and 7 inches high shall be part of the Junction Box. All controls, indicator lights and test points shall be mounted in these two front panels. All other electronic components shall be mounted on the chassis.

3.4.1.3.2 Terminal Boards - Terminal boards fastened to the frame of the Junction Box, and accessible from the front after removal of the chassis, shall provide for all necessary interconnections.

3.4.1.3.3 Controller - A controller shall be provided to supply the Detecting Set, Infrared AN/AAS-15 with I.R. Receiver Head azimuth positioning signals while in the Subsystem search mode.

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3.4.1.3.4 Circuits - Circuits shall be provided for the following functions:

(1) Power control and timing.

(2) Simulation of signals normally provided by the Radar Set AN/APQ-83 or AN/APQ-94.

3.4.1.4 Controls -

3.4.1.4.1 Switches - The following switches shall be provided on the Test Panels:

(1) Circuit breaker - controls 115V 400 cps 3-Phase Power and 28V DC Power.

(2) Pushbutton ON - starts operation of Test Bench.

(3) Pushbutton OFF - stops operation of Test Bench.

(4) Toggleswitch with SUBSYSTEM and SYSTEM position - permits operation with the Radar Test Stand in the System mode or operation without the Radar Test Stand in the Subsystem mode.

(5) Rotary switch, POWER CONTROL.

"OFF" - No DC Power.
"STBY" a) - 28V DC Standby is supplied to Power Supply.
b) - Timing circuit is energized.
c) - 70 sec. delay is started.
d) - 400 cycle excitation is supplied to synchros.
e) - After approximately 70 seconds, 115V 400 cps is applied to Controller to enable auto search capability.

"OPERATE" - If approximately 70 seconds have elapsed since "STANDBY", 28V DC operate will be supplied to Power Supply.

(6) HEAD DRIVE toggleswitch with OFF and NORM positions permits disconnection of power to I.R. Head Az and El Drive motors.

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(7) NUTATION toggleswitch, with OFF and NORM positions, permits disconnection of power to Nutation motor.

(8) CONTROLLER toggleswitch, with OFF and NORM positions, permits disconnection of power to Controller drive motor.

(9) IR MODE toggleswitch, with ON and OFF positions, simulates Radar I.R. Mode selection.

(10) ACTION toggleswitch, in the ACQUISITION and SEARCH-TRACK position, simulates Radar action switch.

(11) Track toggleswitch with TRACK ENABLE and RET TO SEARCH positions enables selection of I.R. Track Mode.

3.4.1.4.2 Indicators - The following indicator lamps shall be provided for visual indication of the various operating modes and logic.

(1) SUBSYSTEM
(2) SYSTEM
(3) POWER ON
(4) CKT BRK ON
(5) STANDBY
(6) OPERATE
(7) ANGLE TRAC K
(8) ACQ. & TRACK
(9) IR MODE
(10) TRACK
(11) IR
(12) SEARCH
(13) LEFT SCAN LIMIT
(14) RIGHT SCAN LIMIT

3.4.1.4.3 Test Points - Test points, to enable external monitoring of various Test Bench Harness and AN/AAS-15 signals, shall be provided as follows:

(1) 115V 400 CPS Phase A
(2) 115V 400 CPS Phase B
(3) 115V 400 CPS Phase C
(4) GROUND (2 provided)
(5) 28V DC RADAR STBY
(6) 28V DC RADAR OPERATE
(7) RCVR DG EXCIT.

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(8) 28V DC POWER ON
(9) GYRO TX EXCIT.
(10) 4 BAR NOD
(11) EL SLAVE ERROR
(12) AZ SLAVE ERROR
(13) DEFL VIDEO
(14) INTENSITY VIDEO
(15) VIDEO RETURN
(16) TONE THRESHOLD
(17) RECEIVER AZ SYNCHRO R1
(18) RECEIVER AZ SYNCHRO R2
(19) RECEIVER AZ SYNCHRO R3
(20) RECEIVER AZ SYNCHRO S1
(21) RECEIVER AZ SYNCHRO S2
(22) RECEIVER AZ SYNCHRO S3
(23) RECEIVER EL SYNCHRO R1
(24) RECEIVER EL SYNCHRO R2
(25) RECEIVER EL SYNCHRO R3
(26) RECEIVER EL SYNCHRO S1
(27) RECEIVER EL SYNCHRO S2
(28) RECEIVER EL SYNCHRO S3

3.4.1.4.4 Aural Tone - A speaker and a phone jack shall be provided for audible monitoring of the aural tone.

3.4.1.5 Electrical Connections - Connections to external circuits shall be provided as shown on HAC Drawing 486902-200.

3.4.2 Rack, Electronic Equipment MT2808/AAM-4, I.R. Receiver - The mounting rack shall provide suitable mounting of the I.R. Receiver and shall permit this unit to be rotated for complete accessibility. An adjustable jack shall be provided as an integral part of the mounting rack to provide for support of the I.R. Target Simulator while being attached to the I.R. Receiver. The mounting rack shall be provided with holes for securing it to the bench top.

3.4.3 Rack, Electronic Equipment MT-2809/AAM-4, Amplifier-Converter - A storage rack for the Amplifier-Converter shall be provided on the upper shelf underneath the bench top.

3.4.4 Rack, Electronic Equipment MT-2810/AAM-4,Power Supply - A storage rack for the Power Supply shall be provided on the lower shelf underneath the bench top.

3 4.5 Extender, Module - Three Module Extenders shall be provided for unit checkout and module fault isolation. One module extender shall accommodate the chassis using the 50 pin connector. A second module extender shall accommodate the chassis using the 25 pin connector. The third module extender shall accommodate the chassis using the 15 pin connector.

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3.4.6 Wrench, Spanner TL-692/AAM-4 - A wrench to facilitate removal of the dome of the Infrared Receiver shall be provided.

3.4.7 Cable Assemblies - Cables shall be provided for the following connections:

(1) From Junction Box to Amplifier-Converter.

(2) From Junction Box to Power Supply.

(3) From Junction Box to I.R. Receiver.

(4) From Junction Box to Liquid Gas Tank.

(5) From Junction Box to Radar Test Harness.

3.4.8 Associated Equipment - The equipment shall operate with the following accessories and these accessories shall not be supplied as part of the equipment.

PN	NOMENCLATURE
AN/AAM-6	Test Set, Infrared Detecting Set
AN/AAM-5	Simulator, Infrared Target
AN/USM-105	Oscilloscope or equivalent

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4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

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4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, and modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181. Drop tests for transit and combination cases shall be in accordance with Specification MIL-T-21200.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. At the discretion of the procuring activity, no production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

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4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction should be used During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing reliability test specified in Para. 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment taken.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (see para. 4.3) the requirement for a special test on an early production equipment may be omitted.

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(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor. The right is reserved by the procuring activity or the government inspector to disapprove, modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

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4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and re-submitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with specification MIL-P-7936.

For level A the method of preservation and packaging of Specification MIL-P-116 for each of the items of the Test Bench Harness shall be as follows:

Items	Type Designation	Applicable Method
Junction Box	J-2121/AAM-4	II
Rack, Electronic Equipment	MT-2808/AAM-4	III
Rack, Electronic Equipment	MT-2809/AAM-4	III
Rack, Electronic Equipment	MT-2810/AAM-4	III
Extender, Module		III
Extender, Module		III
Extender, Module		III
Wrench, Spanner	TL-692/AAM-4	III
Cable Assembly (5)	(*)	III

6. NOTES

6.1 Use - The Test Bench Harness shall be used for maintenance, repair and adjustment of the units of the AN/AAS-15 Infrared Detecting Set. It shall be capable of autonomous operation with the AN/AAS-15 Infrared Detecting Set or it may be operated in a system mode with the Radar Set AN/APQ-83 or AN/APQ-94 and the Radar Test Harness AN/APM-145. The Test Bench Harness shall be mounted on either a Ship Shop Bulkhead Bench or on two suitably modified Shore Shop benches.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC, 400 cycles. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 The parentheses (*) when used in the type designation, shall be replaced by either a number of letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with paragraph 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 48-R308

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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